The `cin` Object

1. Write a program that asks the user for their birth year and then prints their age (assume the current year is 2026). Use `cin` to read the input.

Mathematical Expressions

2. Write a program that calculates the area of a circle. Prompt the user for the radius, then display the area using the formula `area = π * radius²`. Use 3.14159 for π.

Type Conversion

3. What is the output of the following code fragment? Explain why, and show how to fix it to get the expected floating-point result.

```
int a = 7, b = 2;
double result = a / b;
cout << result;
```

Overflow and Underflow

4. On a certain system, the `short` data type can hold values from –32768 to 32767. Assuming two's complement wrap-around behaviour, what will the following program print?

```
short s = 32767;
s = s + 1;
cout << s;
```

Type Casting

5. Write a program that reads two integers from the user and displays their quotient as a decimal number. Ensure that you perform floating-point division by using an explicit type cast (for example, `static_cast<double>()`).

Multiple Assignment and Combined Assignment

6. Rewrite the following statements using combined assignment operators (`+=`, `*=`, `/=`, etc.):

   ```
   x = x + 10;
   y = y * 3;
   z = z / 2;
   total = total + amount;
   ```

   Also write a single statement that sets `a`, `b`, and `c` all to zero using multiple assignment.

Formatting Output

7. Write a program that displays the following table. Use `setw` to align the columns, and use `fixed` and `setprecision` to show exactly two decimal places for the prices.

   ```
   Item        Price
   Apple       1.50
   Bread       2.75
   Milk        3.20
   ```

(Assume the `#include <iomanip>` header is included.)

Working with Characters and `string` Objects

8. Write a program that:

- Reads a user's full name (first name and last name) using `getline`.
- Stores the name in a `string` object.
- Outputs the initials (first character of the first name and first character of the last name) followed by a period.
- Also outputs the total length of the full name string (including the space).

More Mathematical Library Functions

9. Write a program that calculates the hypotenuse of a right triangle. Ask the user for the lengths of the two legs, then use `sqrt` and `pow` from `<cmath>` to compute the hypotenuse and display the result with two decimal places.

Focus on Debugging: Hand Tracing a Program

10. Trace the following program by hand and write its exact output. Assume all necessary headers (`<iostream>`, `<iomanip>`) are included.

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    int x = 5, y = 2;
```

```
    double z = x / y;
    x += 3;
    y *= 2;
    cout << "x=" << x << ", y=" << y << ", z=" << z << endl;
    z = static_cast<double>(x) / y;
    cout << "z now=" << fixed << setprecision(2) << z << endl;
    return 0;
}
```

Focus on Problem Solving: A Case Study

11. **Case Study: Paint Calculator**
    Write a program that calculates the number of gallons of paint needed to paint a rectangular room. The user enters the room's **length**, **width**, and **height** (in feet). The program should:

- Calculate the total area of the four walls (assume no windows or doors) and the area of the ceiling.
- Add the wall area and ceiling area to get the total square footage to be painted.
- Assume one gallon of paint covers **350 square feet**.
- Calculate the required gallons and round **up** to the next whole gallon (use `ceil` from `<cmath>`).
- Display the total square feet and the number of gallons required, formatted neatly with labels and decimal alignment.

    Example input/output (values are examples only):

```
Enter room length (ft): 12
Enter room width (ft): 10
Enter room height (ft): 8

Total area to paint: 536 sq ft
Gallons needed (rounded up): 2
```